

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

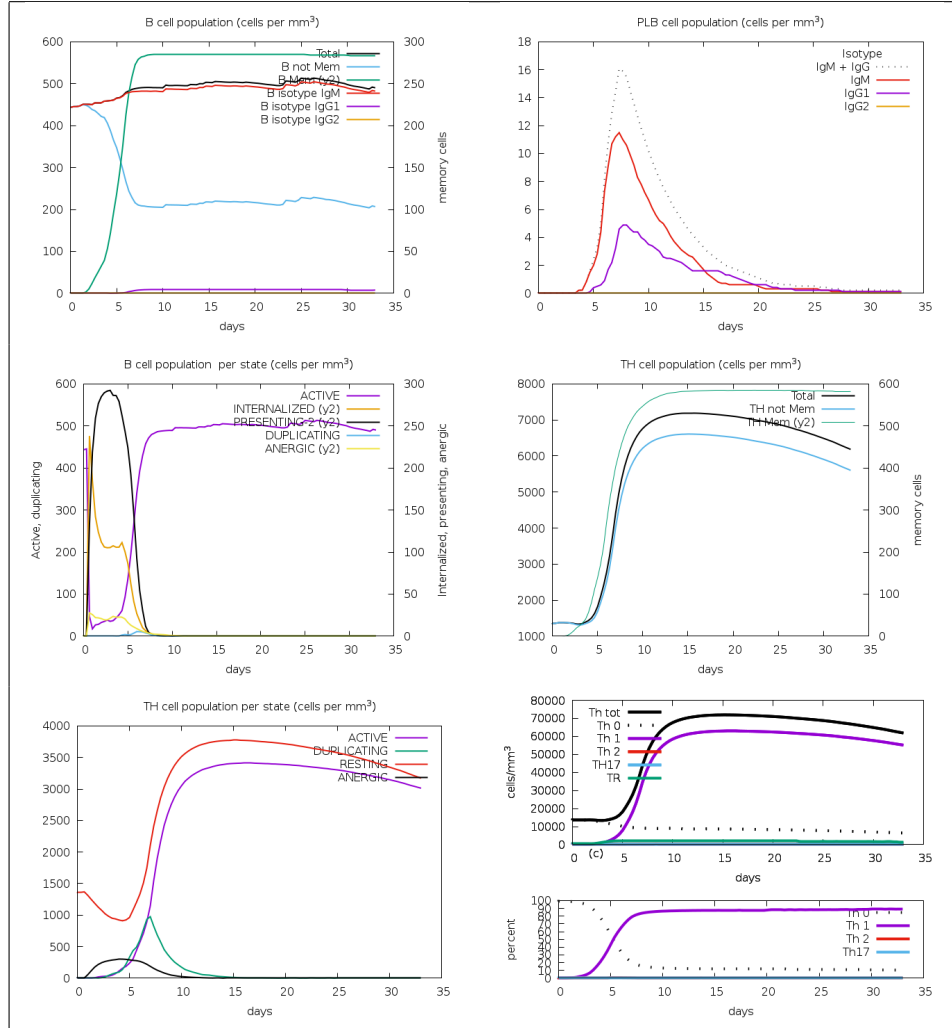


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

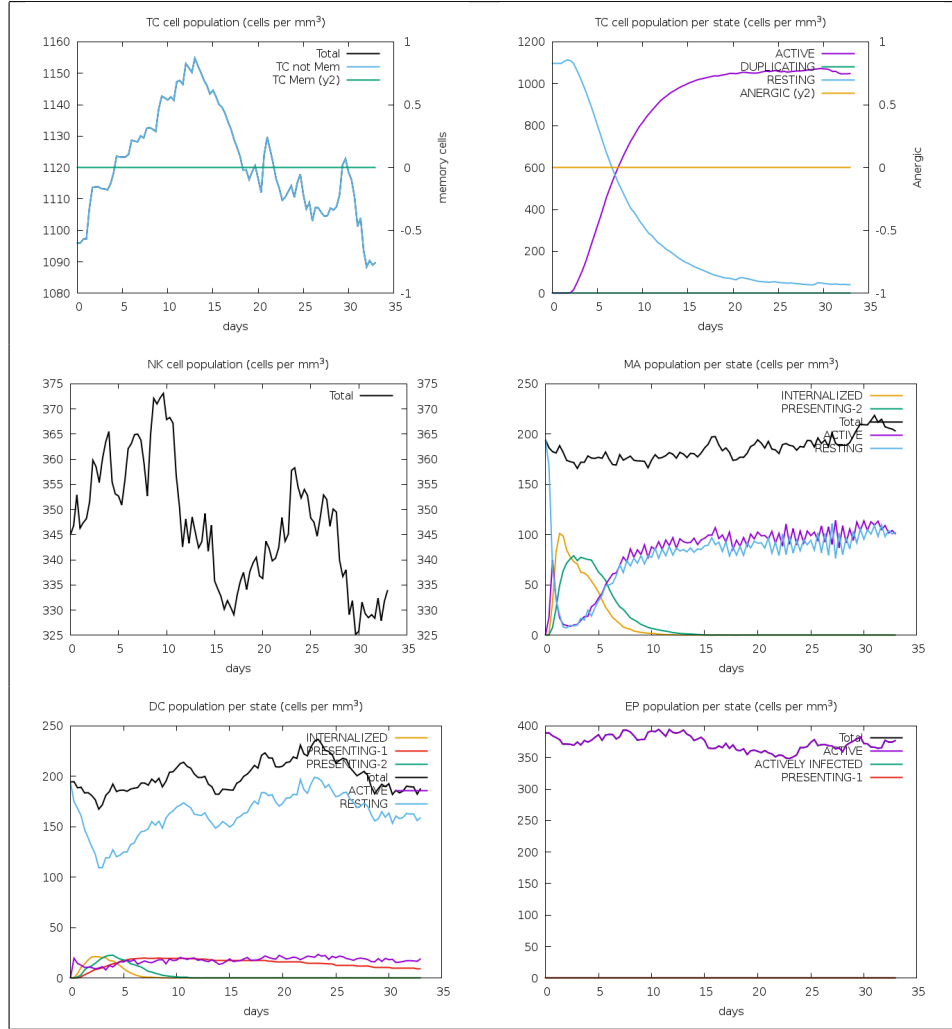


Figure 2: Legend: symbols as figure above.

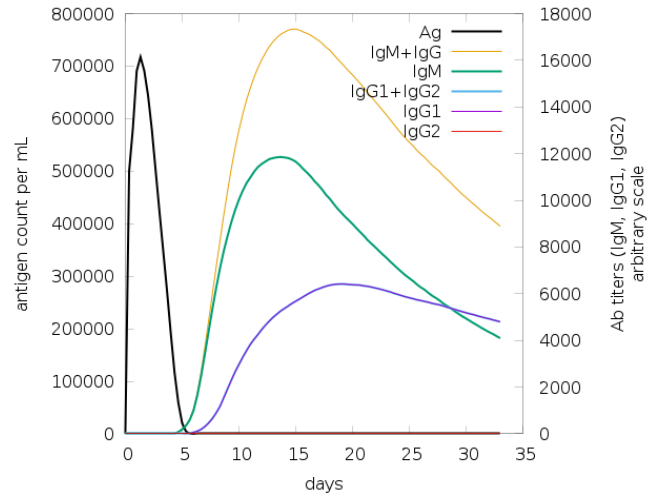


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

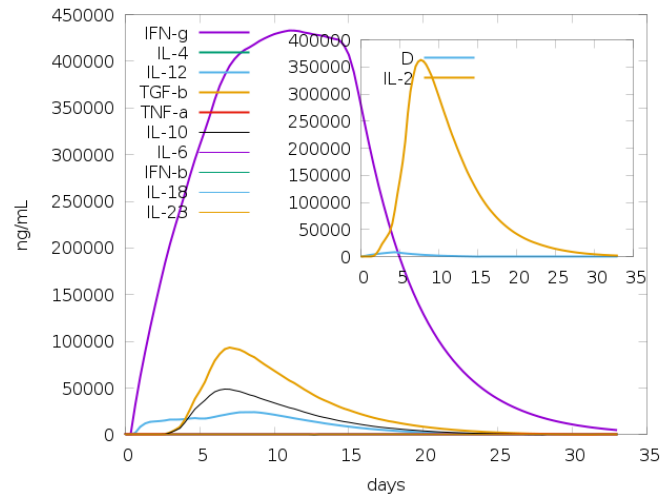


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Use Parker's propensity scale, takes an antigen block as input, and creates a list of residues that are possible epitopes.

APPHALSGGSDQATRVYQDLGGSPQAPEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGG
RPGAPGSGSGSPRHRDGVRRPQKRPSCIGCKGTHGTTGAGAGAGGAGAGAGGAGAGGGAGGAGGAGAGGGGAGA
GGGAGGAGGAGAGGGAGAEAAAKRVLVSVVLLGPGPGTMEFVSVAFGPGPGACMLAPLEKGPGPGLPQQQLTAYGPGPG
FLDKGTYTLGPGPGVFSDEIPASQDMPTNTEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRSALYEAST

[illegible]

```

1]      pos=7 len=6      GGSDQA
2]      pos=23 len=4     SQAP
3]      pos=32 len=6     GPGTGP
4]      pos=41 len=53     LGEKGDTSGPSGGSGPQRRGGDNHGRGRGRGRGGRRPGAPGSGSGPRH
5]      pos=95 len=4      DGVR
6]      pos=111 len=66    GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGAGGGAGAGGGAGGAGAGGGAG
7]      pos=264 len=4     DMPT
8]      pos=282 len=19    AAEGDDGDGPGPGSDREVR

```

Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 4 MHCI molecules

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

APPHALSGGSDQATRVYQDLGGPSQAPAEAAAKGPGTGPGNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGG
RPGAPGSGSGSPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGAGAGGGAGAGGGAGGAGAGGGAGAGGGAGA
GGGAGGAGGAGAGGGAGAEAAAKRVLVSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGPGLPQQQLTAYGPGPG
FLDKGTYTLGPGPGVFSDEIPASQDMPNTAEAAKPPMVEGAAAEAGDDGDGPGPGSDREVRSALYEAST

```
0] pos= 8 score=0.052333 unnormalised=5.7896000000 GSDQATRVY
1] pos= -1 score=0.947667 unnormalised=104.8410000000 non-binding event
```

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-011134_5_aLY24ET6omnf.FSA_1_001

APPHALSGGSDQATRVYQDLGGPSQAPEAAAKPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGG
RPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGAGGAGAGGGAGA
GGGAGGAGGAGAGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQLTAYGPGPG
FLDKGTYTLGPGPGVFSDEIPASQDMPTNTEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALYEAST

Epitopes of protein 0 -----

0]	pos=	8	score=0.052333	unnormalised=5.7896000000	GSDQATRVY
1]	pos=	-1	score=0.947667	unnormalised=104.8410000000	non-binding event

=====
Allele: B0702
Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-011134_5_aLY24ET6omnf.FSA_1_001

APPHALSGGSDQATRVYQDLGGPSQAPEAAAKPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGG
RPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGAGGAGAGGGAGA
GGGAGGAGGAGAGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQLTAYGPGPG
FLDKGTYTLGPGPGVFSDEIPASQDMPTNTEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALYEAST

Epitopes of protein 0 -----

0]	pos=	21	score=0.020802	unnormalised=2.4282000000	GPSQAPEAA
1]	pos=	80	score=0.002580	unnormalised=0.3012000000	RPGAPGGSG
2]	pos=	99	score=0.016553	unnormalised=1.9322000000	RPQKRPSCI
3]	pos=	223	score=0.013426	unnormalised=1.5672000000	GPGLPQGQL
4]	pos=	259	score=0.017444	unnormalised=2.0362000000	IPASQDMPT
5]	pos=	265	score=0.031023	unnormalised=3.6212000000	MPTNTEAAA
6]	pos=	-1	score=0.898171	unnormalised=104.8410000000	non-binding event

=====
Allele: B0702
Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-011134_5_aLY24ET6omnf.FSA_1_001

APPHALSGGSDQATRVYQDLGGPSQAPEAAAKPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGG
RPGAPGGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGAGGAGAGGGAGA
GGGAGGAGGAGAGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQGQLTAYGPGPG
FLDKGTYTLGPGPGVFSDEIPASQDMPTNTEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALYEAST

Epitopes of protein 0 -----

0]	pos=	21	score=0.020802	unnormalised=2.4282000000	GPSQAPEAA
1]	pos=	80	score=0.002580	unnormalised=0.3012000000	RPGAPGGSG

4]	pos= 218	score=0.010733	unnormalised=0.8915600000	LEKGP GPGL
5]	pos= 246	score=0.088126	unnormalised=7.3205600000	YTLGP GPV
6]	pos= -1	score=0.796539	unnormalised=66.1680000000	non-binding event
